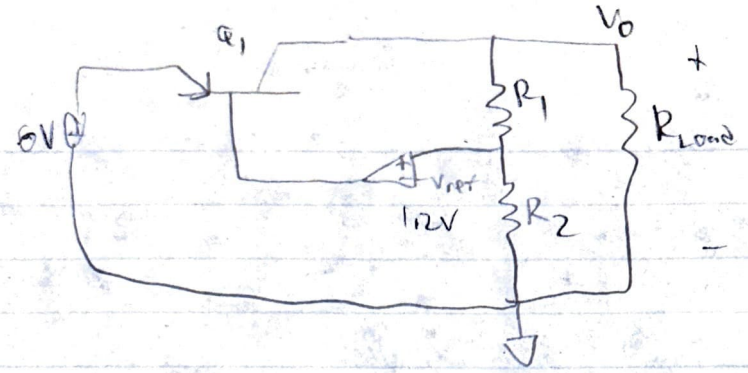
ECEN 164 Prelab 5



Ideal devices

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a. $\frac{3.3}{6} = \boxed{55\% \text{ eff}}$

b. $\frac{3.3}{50} = \boxed{66\text{ mA} \text{ needs to be supplied by vm}}$

c.

i. To stabilize V_o and make ev voltage input. (V_{ref})

ii.

$$3.3\left(\frac{150K}{R1+150K}\right) = 1.2 \rightarrow (R1+150K)\frac{1.2}{3.3} = 150K$$

$$\boxed{R1 = 262.5k\Omega}$$

iii. No it actually helps regulation

d. It's where the V_{BE} drop is. On the PNP, it won't disrupt V_o and thus we can have a higher Z_o.

e. We want a MOS b/c it has (virtually) no ground current and thus is more eff.